

Same Accuracy, Different Geometry: AdamW vs. Muon for Fine-Tuning BERT-small on SST-2

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Goal and setup. I fine-tune BERT-small (4 layers, 29M parameters) on SST-2 with AdamW and with Muon, asking not only which optimizer is better but how their optimization differs. The training and evaluation loop is my own; libraries only load the model, tokenizer and data. All runs use batch size 32, maximum length 64, 6% warmup followed by linear decay, weight decay 0.01 and no gradient clipping. The full study used about 50 minutes of training compute on a single T4 GPU.

Fair comparison.

- **One difference.** Only the update rule for the 25 hidden weight matrices (attention, feed-forward, pooler) differs. Embeddings, biases, LayerNorm and the classifier use AdamW with identical settings in both runs, which is how Muon is meant to be used.
- **Comparable step sizes.** Muon orthogonalizes the momentum matrix ($M = U\Sigma V^\top \mapsto UV^\top$, five Newton-Schulz steps) and scales the result by $0.2\sqrt{\max(m, n)}$, so its update RMS matches AdamW’s and both optimizers can share one learning-rate grid.
- **Equal tuning.** Both get the same one-epoch sweep over $\{3\cdot 10^{-5}, 10^{-4}, 3\cdot 10^{-4}, 10^{-3}\}$ with a separate seed, selected by accuracy on 2,000 held-out training examples. Both chose 10^{-4} .
- **Paired runs.** Final runs train for 2 epochs (4,084 steps) with 3 seeds. Each seed fixes the classifier initialization and data order for both optimizers, so every comparison is paired.
- **Clean test set.** SST-2 test labels are hidden, so the official validation set (872 sentences) serves as the test set and is never used for any decision.

Metrics. The table lists each metric and the question it answers. Outcome metrics say which optimizer did better; behavior metrics say how each one moves through weight space; geometry metrics describe the solution it reaches. Sharpness is measured at the final weights on 512 fixed training examples, with dropout off, over all non-embedding parameters. I use four estimators because each has a blind spot. The top Hessian eigenvalue $\lambda_{\max}$ and the trace $\text{tr}(H)$, computed from exact Hessian-vector products, change under weight rescalings that leave the function unchanged. Random sharpness along filter-normalized directions and adaptive worst-case sharpness are normalized by weight size. All four shrink as predictions grow more confident, so confidence is reported as a control.

Metric	What it reveals	AdamW	Muon
<i>Outcome</i>			
Test accuracy (%)	task performance	89.07 ± 0.40	88.69 ± 0.93
Test loss	finer-grained than accuracy on a small test set	0.333 ± 0.011	0.372 ± 0.010
Final training loss	how well the optimizer fits the training data	0.124 ± 0.009	0.107 ± 0.008
Steps to 93% val. accuracy	convergence speed	2083 ± 144	1417 ± 144
Time per step (ms)	computational cost	45.7	105.5
<i>Optimizer behavior</i>			
Update size $\ \Delta W\ _F / \ W\ _F (\times 10^{-4})$	checks that step sizes are matched	2.43 ± 0.06	2.03 ± 0.00
$\cos(\Delta W, -G)$	how far the update departs from the gradient direction	0.283 ± 0.001	0.113 ± 0.002
$\ W - W_0\ _F / \ W_0\ _F$	how far training moves from the pretrained weights	0.075 ± 0.002	0.102 ± 0.001
Stable rank of $W - W_0$	how many directions the total change uses (max 512)	10.0 ± 0.1	111 ± 2
<i>Solution geometry (lower sharpness is flatter)</i>			
Confidence on probe set	control: more confident models look flatter	0.972 ± 0.001	0.983 ± 0.001
$\lambda_{\max}(H)$	worst-case curvature	16.7 ± 0.7	28.0 ± 3.2
$\text{tr}(H)$	average curvature	166 ± 47	251 ± 116
Random sharpness, $\alpha = 0.5$	loss rise along random, scale-normalized directions	0.081 ± 0.004	0.278 ± 0.004
Worst-case sharpness, $\rho = 0.2 (\times 10^{-3})$	largest loss rise in a scale-normalized ball	4.4 ± 0.3	5.9 ± 0.5
Barrier between solutions	whether both solutions lie in one basin	0.014 ± 0.004	

Mean $\pm$ standard deviation over 3 paired seeds. Curves over training and landscape plots are in Figure 1.

Which optimizer performed better? By test accuracy, neither: Muon minus AdamW is -0.38 ± 1.10 points (paired t -test, $p = 0.61$), below the ≈ 0.45 -point standard error of a single evaluation on 872 sentences. By test loss, AdamW is better in every seed (0.039 ± 0.002 , $p < 0.001$), and it is $2.3\times$ faster per step. Muon fits the training data faster and further (Fig. 1a, b), but its test-train loss gap is larger (0.26 vs. 0.21), so the extra fit does not generalize. Muon also tolerates a larger learning rate: at 10^{-3} AdamW failed (56% validation accuracy) while Muon reached 93%. Step sizes are matched

(Fig. 1d), so the difference is in direction. Muon’s total change uses $11\times$ more directions (Fig. 1f), moves 37% farther from the pretrained weights (Fig. 1c) and is less aligned with the gradient (Fig. 1e), while AdamW’s change is low-rank.

Which solution is flatter? AdamW’s, by all four estimators. $\lambda_{\max}$ and worst-case sharpness do not overlap across seeds, the trace agrees but is noisy, and the random-direction gap is clear at larger radii (Fig. 1h). The result is conservative: Muon’s solution is more confident, which lowers curvature and should make it look flatter, not sharper. During training, Muon’s $\lambda_{\max}$ first rises from 110 to 149 before falling, while AdamW’s decreases steadily (Fig. 1g). The loss barrier between the two solutions is small, so they lie in one basin, and their average has lower test loss than either one (Fig. 1i).

How reliable is this? The structural, geometric and test-loss differences hold in every seed and are large relative to seed variation, so I consider them fairly robust. The accuracy comparison is underpowered (3 seeds, 872 sentences). Learning rates were chosen with one seed on small margins, and Muon at $3\cdot 10^{-4}$ had lower validation loss, so Muon may be slightly under-tuned. The held-out split is also optimistic (94 to 95% vs. 88 to 90% on test) because SST-2 training phrases overlap. Sharpness is measured at one point and in Euclidean geometry, not Muon’s natural spectral geometry. Muon starts from AdamW-pretrained weights, its timing reflects unoptimized code, and the link between sharpness and generalization here is only correlational.

What would I change in a larger study? Tune learning rate, momentum and weight decay jointly, selecting by validation loss; use 10+ seeds with confidence intervals, several tasks and model sizes, and an out-of-distribution test set; compare at equal wall-clock time with an optimized Muon; track sharpness throughout training, at matched training loss and in each optimizer’s own geometry; and extend the study to SFT and RLVR.

Problems encountered.

- Hessian-vector products need second derivatives, which fused attention kernels do not support. I switched to standard attention, verified the products against finite differences, and checked that Newton-Schulz pushes singular values close to 1.
- AdamW and Muon learning rates are not directly comparable. The RMS scaling solved this, and the matched update sizes confirm it.
- Hidden test labels forced model selection onto a held-out training split, which turned out to be optimistic.

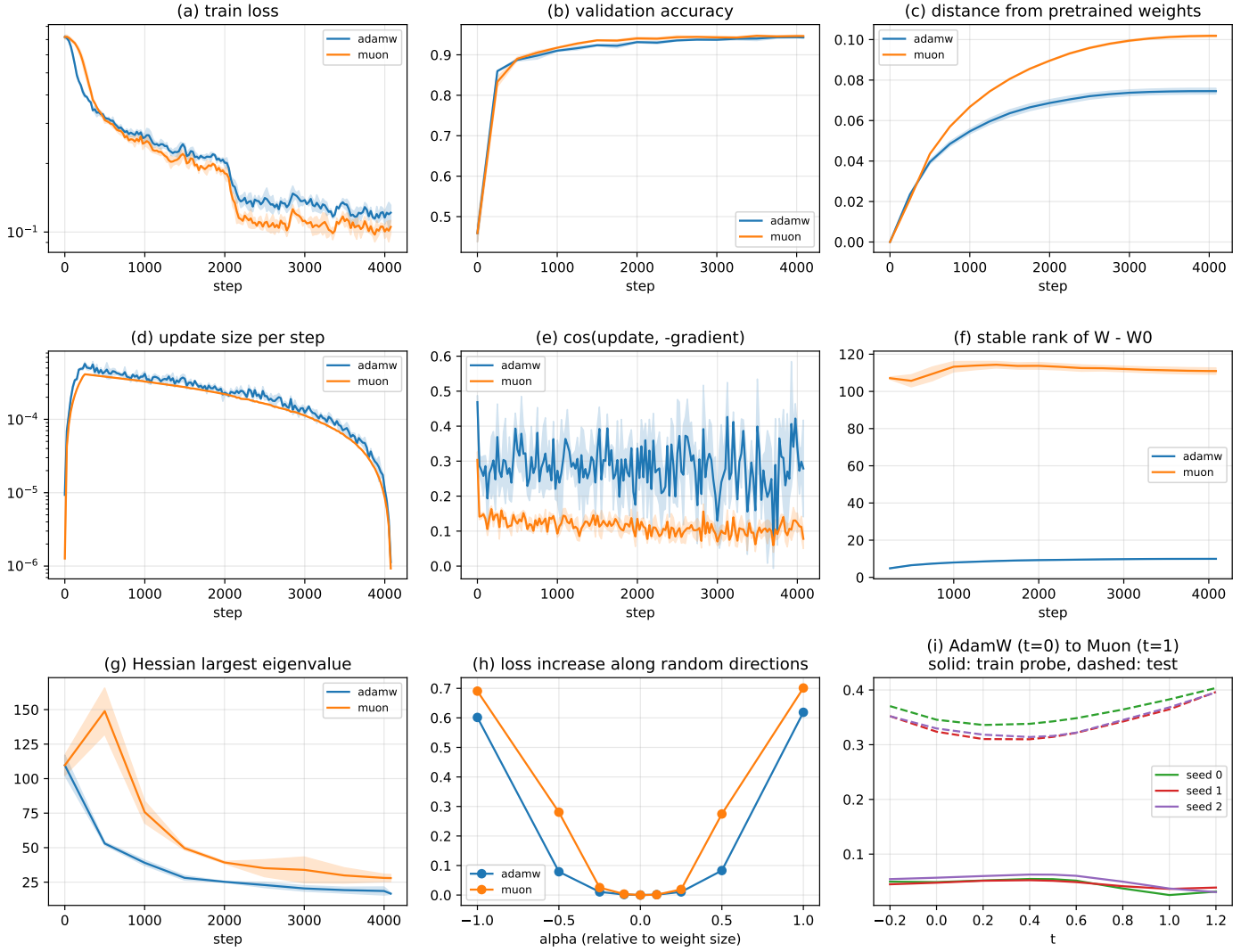


Figure 1: Training dynamics and solution geometry, mean $\pm$ std over 3 seeds. (a) Training loss. (b) Validation accuracy: Muon reaches 93% about 670 steps earlier. (c) Distance from the pretrained weights: Muon travels 37% farther. (d) Update size per step: matched by construction, confirming a fair step size. (e) Alignment with the negative gradient: Muon departs much further from the gradient direction. (f) Stable rank of $W - W_0$: Muon’s change uses about 111 directions from the start, AdamW’s about 10. (g) Top Hessian eigenvalue: AdamW’s falls steadily, Muon’s first rises, then settles higher. (h) Loss increase along filter-normalized random directions: the Muon solution is sharper at every radius beyond 0.1. (i) Loss on the line from the AdamW solution ($t = 0$) to the Muon solution ($t = 1$); solid: training probe set, dashed: test set. The training-loss barrier is small, and the test loss is lowest between the two solutions.